

 NAACL 2025 | Paper #861

AdTEC: A Unified Benchmark for Evaluating Text Quality in Search Engine Advertising

Peinan Zhang, Yusuke Sakai, Masato Mita, Hiroki Ouchi, Taro Watanabe

 CyberAgent®



NAIST



CyberAgent **AI Lab**



2408.05906



CyberAgentAILab/AdTEC



cyberagentailab.github.io/AdTEC

Motivation

Online advertising is a huge industrial use case for NLP

Market size

Worldwide digital advertising spending is expected to reach \$723B ^[1] in 2026

Motivation

Online advertising is a huge industrial use case for NLP

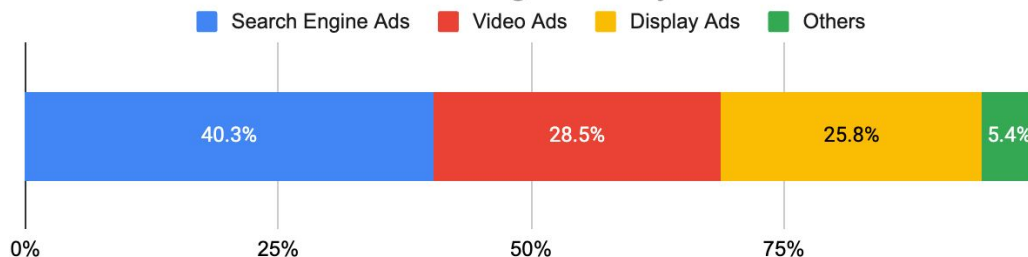
Market size

Worldwide digital advertising spending is expected to reach \$723B [1] in 2026

Share of Costs by Ad Format and Text Information [2]

- Search Engine Ads: 40.3%, 100%
- Video Ads: 28.5%, 25%
- Display Ads: 25.8%, 25%

Distribution of Online Advertising Costs by Ad Format



Motivation

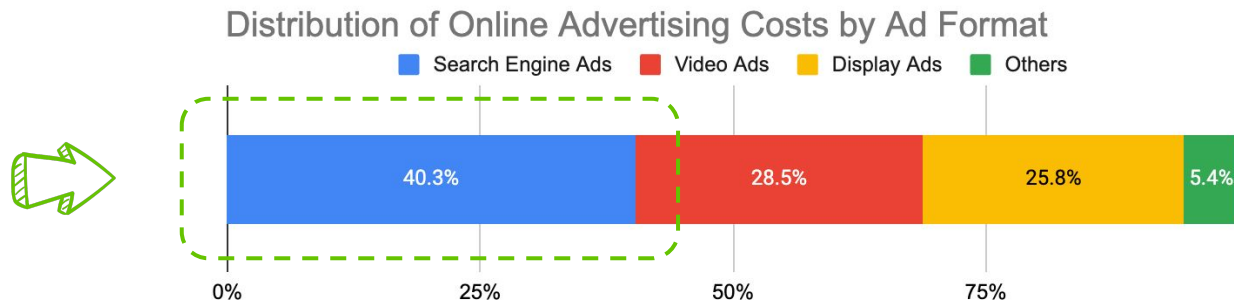
Online advertising is a huge industrial use case for NLP

Market size

Worldwide digital advertising spending is expected to reach \$723B [1] in 2026

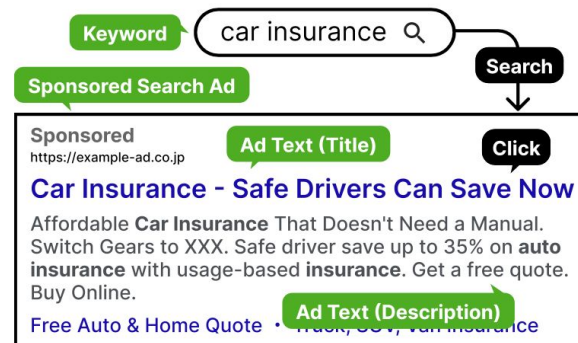
Share of Costs by Ad Format and Text Information [2]

- **Search Engine Ads: 40.3%, 100%**
- Video Ads: 28.5%, 25%
- Display Ads: 25.8%, 25%



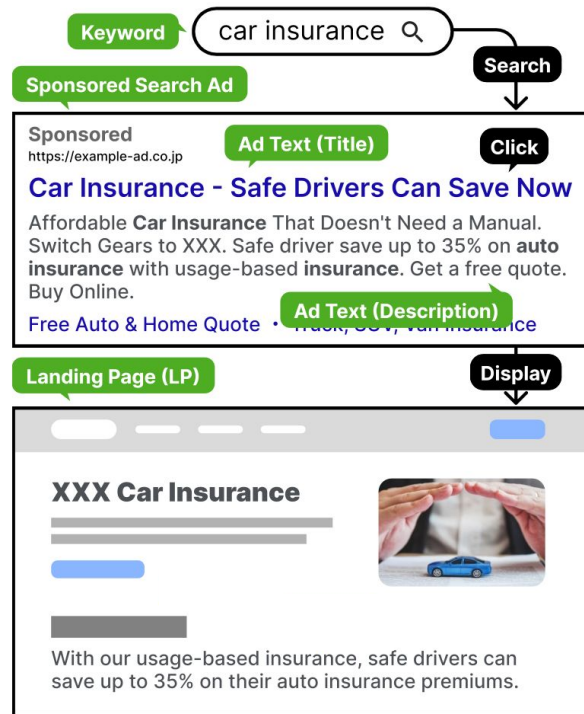
Search Engine Ads a.k.a Sponsored Search Ads

- 1 After customers enter **keywords** into search engines, **titles** and **descriptions** of relevant ads appear as part of the search results.



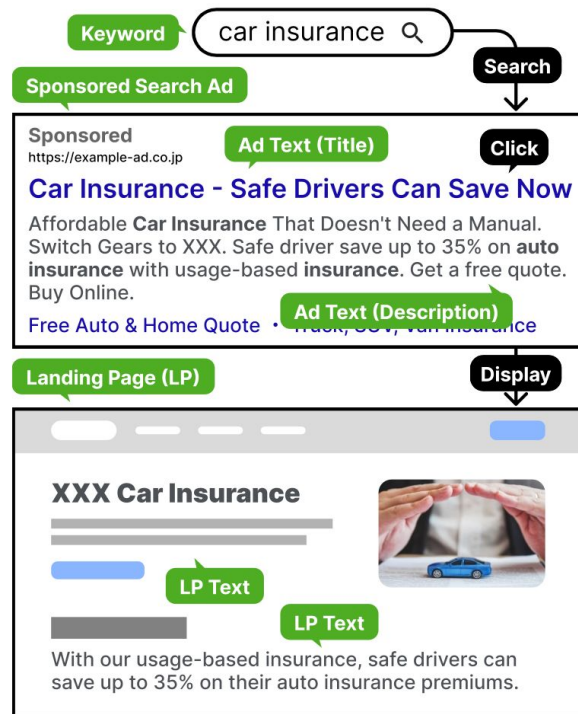
Search Engine Ads a.k.a Sponsored Search Ads

- 1 After customers enter **keywords** into search engines, **titles** and **descriptions** of relevant ads appear as part of the search results.
- 2 When a customer clicks an ad's URL, they are directed to a web page known as a **landing page (LP)**.



Search Engine Ads a.k.a Sponsored Search Ads

- 1 After customers enter **keywords** into search engines, **titles** and **descriptions** of relevant ads appear as part of the search results.
- 2 When a customer clicks an ad's URL, they are directed to a web page known as a **landing page (LP)**.
- 3 The LP contains **texts** and **images** related to the advertiser's products, prompting the customer to view the product or make purchase decisions.



An era of mass production of ad creatives in academia

Generating Better Search Engine Text Advertisements with Deep Reinforcement Learning

J Weston Hughes*
UC Berkeley
Berkeley, CA
jwhughes@berkeley.edu

Keng-hao Chang
Microsoft
Sunnyvale, CA
kenchan@microsoft.com

Ruofei Zhang
Microsoft
Sunnyvale, CA
bzhang@micros

← KDD 2019 [3] NAACL 2021 [4] →

An Empirical Study of Generating Texts for Search Engine Advertising

Hidetaka Kamigaito^{*1}, Zhang Peinan^{*2}, Hiroya Takamura³ and Manabu Okumura¹

¹Tokyo Institute of Technology, ²CyberAgent, Inc.
Institute of Advanced Industrial Science and Technology (AIST)
r.pi.titech.ac.jp, zhang_peinan@cyberagent.co.jp
a.hiroya@aist.go.jp, oku@lr.pi.titech.ac.jp

DeepGen: Diverse Search Ad Generation and Real-Time Customization

Konstantin Golobokov[◇], Junyi Chai[★], Victor Ye Dong[★],
Mandy Gu[★], Bingyu Chi[★], Jie Cao[★], Yulan Yan[★], Yi Liu[★]
◇ Azure AI, ★ Bing Ads, ★ News & Feeds
Microsoft, Redmond, USA
{FirstName.LastName}@microsoft.com

Striking Gold in Advertising: Standardization and Exploration of Ad Text Generation

Masato Mita, Soichiro Murakami, Akihiko Kato, Peinan Zhang

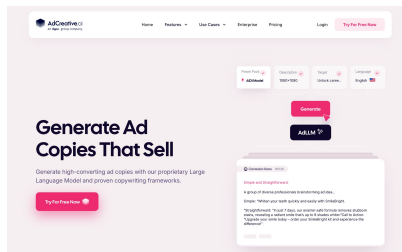
CyberAgent.

{mita_masato, murakami_soichiro, kato_akihiro, zhang_peinan}@cyberagent.co.jp

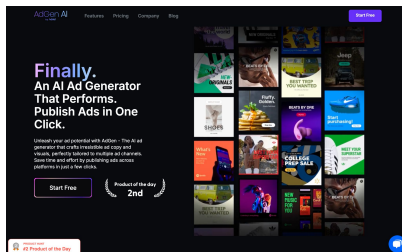
EMNLP 2022 [5] ACL 2024 [6] →

An era of mass production of ad creatives **in the industry**

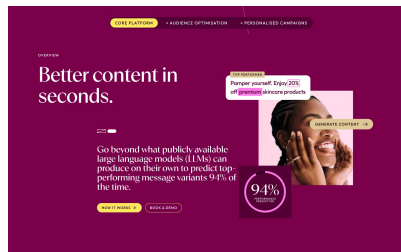
An increasing number of services are emerging to mass-produce ads using generative AI technology.



AdCreative.ai by Appier [8]



AdGen AI by Adriel [9]



JACQUARD [10]



KiwamiTD by CyberAgent [11]



広告クリエイティブを革新する「∞AI（ムゲンエーアイ）」を開発
電通デジタルは、デジタル広告の運用型広告において、広告クリエイティブ…

Mugen AI by Dentsu [12]



博報堂D Yホールディングス、博報堂テクノロジーズ統合マーケ…
—AIの活用で検索連動型広告における配信業務の効率化・高度化を推進—

CREATIVE BLOOM TEXT Ads
by Hakuodo [13]

Problem

How can we ensure the quality of
mass-produced creatives?

Problem

How can we ensure the quality of
AdTEC
mass-produced creatives?

AdTEC is ...

Ad Text Evaluation Benchmark by **CyberAgent**

... **the first public dataset** designed for
evaluating the quality of ad texts
based on **real-world AdOps workflows.**

Main Contributions

Task design and Dataset construction

We **organized advertising workflows** based on real-world advertising operations (AdOps), **designed five related tasks**, and **created datasets** to evaluate ad text quality based on the workflow. We **packaged them as a benchmark called AdTEC**.

Experiment and Analysis

We **validated the performance of existing PLMs and human evaluators** on AdTEC, analyzed the characteristics, **demonstrated that our benchmark is challenging**, and **highlighted potential areas for improvement** and future research.

Main Contributions

Task design and Dataset construction

We **organized advertising workflows** based on real-world advertising operations (AdOps), **designed five related tasks**, and **created datasets** to evaluate ad text quality based on the workflow. We **packaged them as a benchmark called AdTEC**.

Experiment and Analysis

We **validated the performance of existing PLMs and human evaluators** on AdTEC, analyzed the characteristics, **demonstrated that our benchmark is challenging**, and **highlighted potential areas for improvement** and future research.

Main Contributions

Task design and Dataset construction

We **organized advertising workflows** based on real-world advertising operations (AdOps), **designed five related tasks**, and **created datasets** to evaluate ad text quality based on the workflow. We **packaged them as a benchmark called AdTEC**.

Experiment and Analysis

We **validated the performance of existing PLMs and human evaluators** on AdTEC, analyzed the characteristics, **demonstrated that our benchmark is challenging**, and **highlighted potential areas for improvement** and future research.

Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required

Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground

Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground

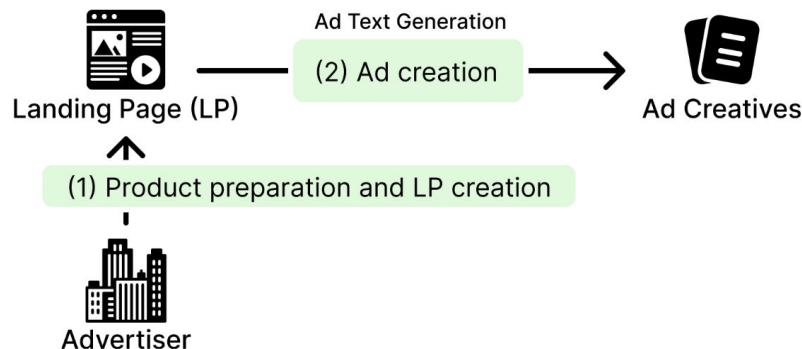


(1) Product preparation and LP creation



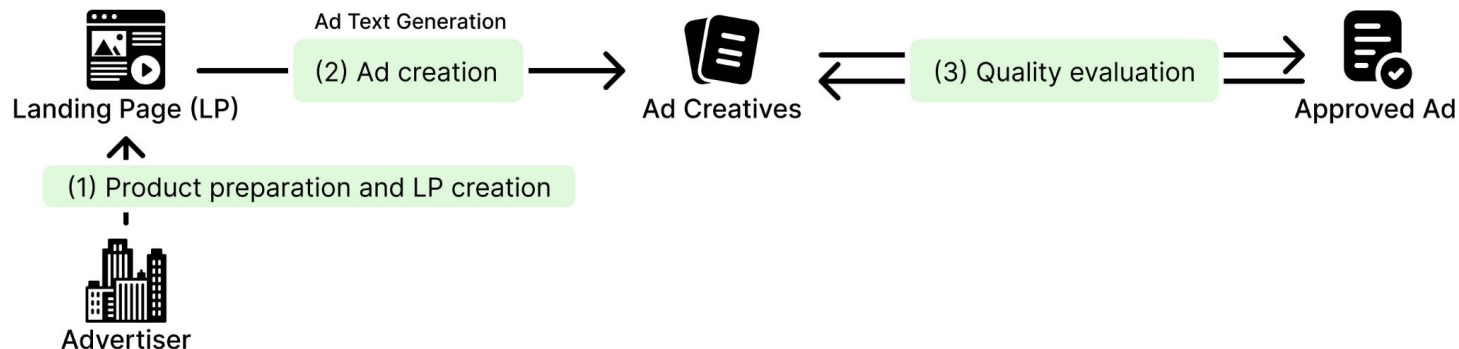
Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



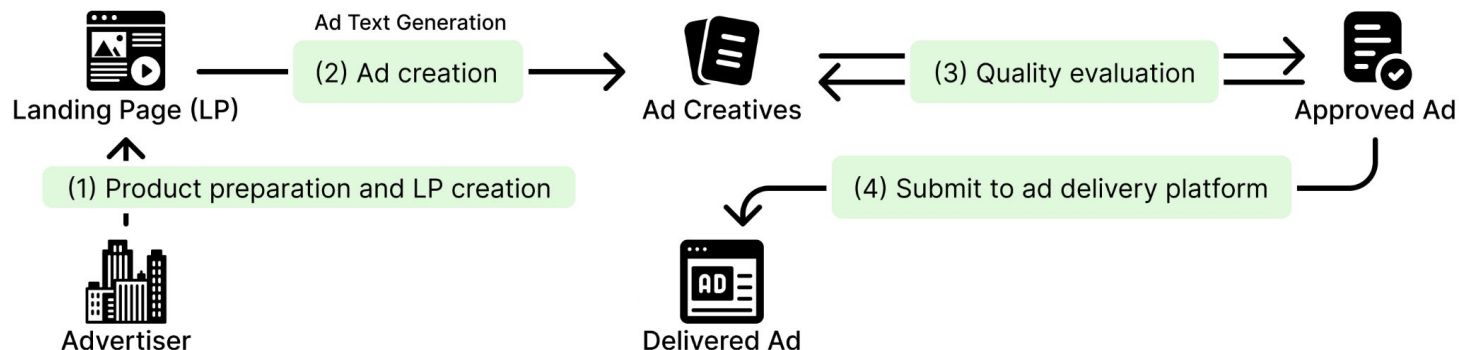
Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



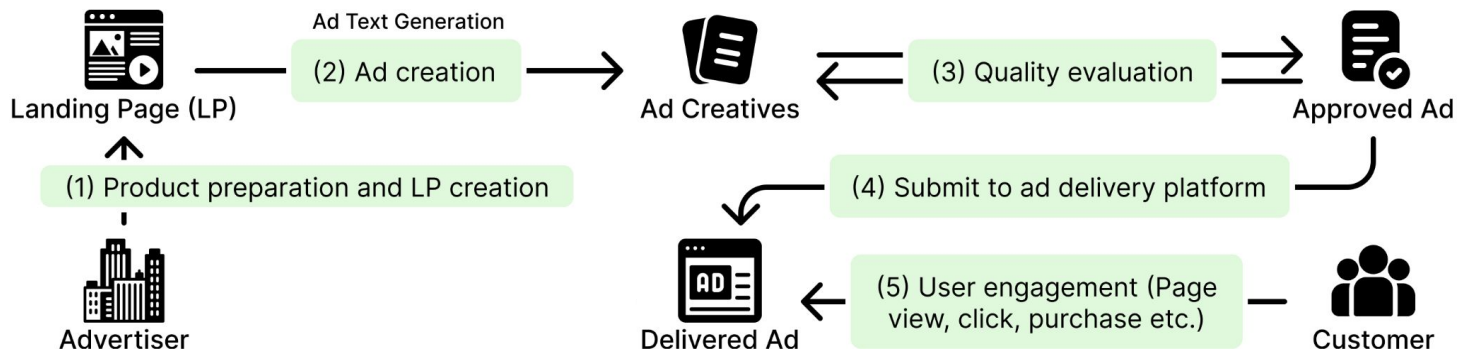
Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



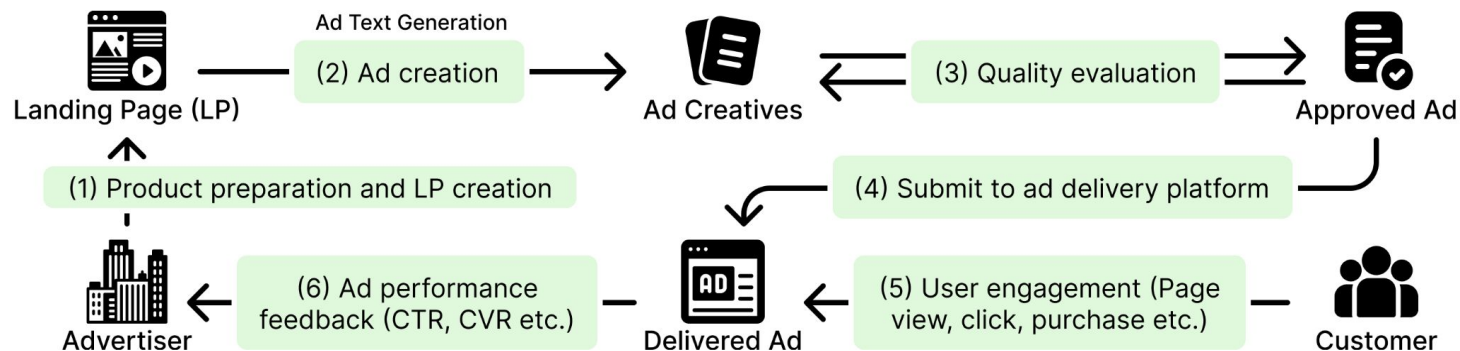
Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



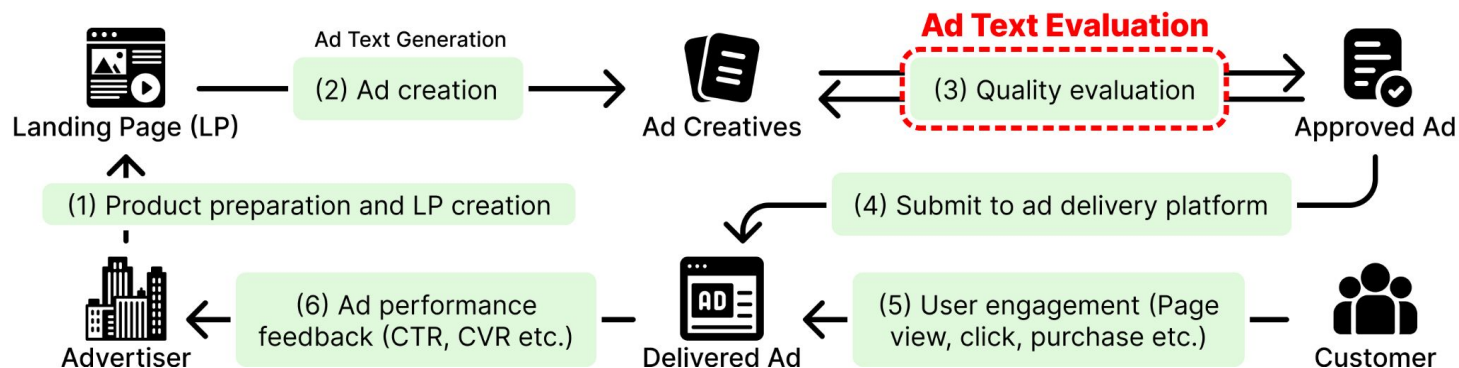
Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



Organizing Advertising Workflows

- **Advanced expertise is required for AdOps**
 - a detailed understanding of the delivery infrastructure, characteristics, formats, and etc. is required
- **Interviews with two types of experts in advertising operation and generalized workflow**
 - those who oversee the operations department of an advertising agency
 - those who are directly involved in operations on the ground



Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Ad Text 1

Condominium Sales /
Free Assessment Now



Ad Text 2

Engineer's career /
Engineer's Job



unnatural repetition

Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Ad Text 1

Condominium Sales /
Free Assessment Now

LP Text 1

Let XX assess your
single-family home!



inconsistent:
condominium vs. single-family home

Ad Text 2

Engineer's career /
Engineer's Job

LP Text 2

Find jobs at website
XX!



Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Title [No.1] Card loan comparison site

Description The best place to get a card loan without telling anyone. [...]

Keyword card loan

Industry finance



Quality score: 82.3

Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Ad Text

Affordable Car Insurance That Doesn't Need a Manual!

Switch gears to XXX. Safe driver save up to 35% on auto insurance with usage-based insurance. Get a free quote. Buy Online.



Features: "That Doesn't Need a Manual"

Special deals: "save up to 35%", "Get a free quote"

User-friendliness: "Buy Online"

Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Ad Text 1a

Suppon Black Vinegar with
Luxury Ceramide

Ad Text 1b

Suppon Black Vinegar and
Luxury Ceramide



5.0

Ad Text 2a

Find a gift that fits your
budget

Ad Text 2b

Save up to 40% on
discounted products



2.3

Task Design



Ad Acceptability

Predicting the overall quality



Ad Consistency

Predicting the consistency between LP content and ad text



Ad Performance Estimation

Estimating the quality score based on ad delivery information



A3 (Aspect of Ad Appeal) Recognition

Recognizing the entities of A3s in an ad text [14]



Ad Similarity

Predicting the similarity of a pair of ad texts

Dataset Construction

Data Collection

- Data from actual AdOps workflow in CyberAgent
- Delivered ads between 2021 and 2022
- All data are in Japanese

Data Pre-processing

- Deduplication
- Masking the sensitive entities
- Pseudo-similar pairs based on the ad hierarchical structure

Data Splitting

- Based on the ad hierarchical structure

Dataset Construction

Data Collection

- Data from actual AdOps workflow in CyberAgent
- Delivered ads between 2021 and 2022
- All data are in Japanese

Data Pre-processing

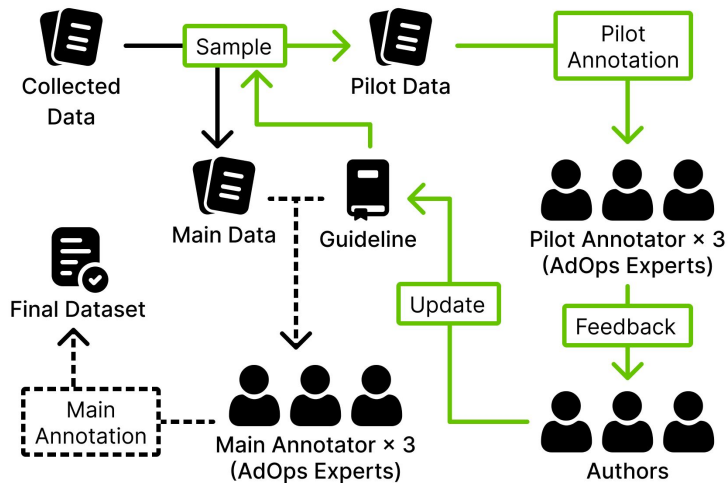
- Deduplication
- Masking the sensitive entities
- Pseudo-similar pairs based on the ad hierarchical structure

Data Splitting

- Based on the ad hierarchical structure

Note: Preprocessing, data splitting, and annotation vary depending on the task. Please refer to the paper for details.

Annotation Workflow



Green line: Guideline Creation (repeated twice)

Dashed line: Main Annotation (only once)

Overview of the Final Dataset

Task	Train/Dev/Test	Setup: Input $\rightarrow$ Label	Metrics
Ad Acceptability	13265/970/980	Classification: Ad text $\rightarrow$ acceptable/unacceptable	Accuracy/F1 Score
Ad Consistency	10635/945/970	Classification: (Ad text, LP text) $\rightarrow$ consistent/inconsistent	Accuracy/F1 Score
Ad Performance Estimation	125087/965/965	Regression: (Ad texts, Keyword, Industry) $\rightarrow$ [0, 100]	Pearson/Spearman corr.
A3 Recognition	1856/465/410	Classification: Ad text $\rightarrow$ multi-labels	F1-micro/-macro
Ad Similarity	4980/623/629	Regression: Ad text pair $\rightarrow$ [1, 5]	Pearson/Spearman corr.

Experiment

Baseline Models

Performance of Pre-trained Language Models (PLMs) and Human Evaluators

- **Encoder PLMs with fine-tuning**
 - Commonly employed BERT-base models (Tohoku BERT, Waseda BERT, XLM-RoBERTa)
- **Large Language Models (LLMs) with In-Context Learning (ICL)**
 - Open-sourced LLMs (CALM2-7b, ELYZA-7b) and Private LLM (GPT-4)
- **LLMs with fine-tuning**
 - ELYZA-7b
- **Human Evaluators**
 - Three human evaluators not involved in the annotation phase
 - Same workflow with annotations

Experiment

Quantitative Results

Evaluator	AD ACCEPT. Accuracy/F1-score	AD CONSIST. Accuracy/F1-score	AD PERFORM. EST. Pearson/Spearman	A ³ RECOGNITION F1-micro/-macro	AD SIMILARITY Pearson/Spearman
<i>Fine-tuned Encoder Models</i>					
Tohoku BERT _{BASE}	0.685/0.691	0.757/0.504	0.437/0.454	0.753/0.629	0.769/0.803
Tohoku BERT _{LARGE}	0.711/0.688	0.767/0.552	0.480/0.497	0.774/ 0.694	0.773/0.807
Waseda BERT _{BASE}	0.615/0.639	0.725/0.388	0.444/0.454	0.641/0.442	0.749/0.797
Waseda BERT _{LARGE}	0.598/0.637	0.755/0.474	0.445/0.457	0.663/0.517	0.740/0.800
XLM-RoBERTa _{BASE}	0.694/0.677	0.743/0.465	0.425/0.439	0.730/0.542	0.846/0.870
XLM-RoBERTa _{LARGE}	0.705/0.690	0.758/0.519	0.453/0.457	0.778/0.677	0.878/0.878
<i>Zero-/Few-shot LLMs</i>					
CALM2 _{7b}	0.520/0.115	0.381/0.472	0.006/0.013	0.154/0.042	0.036/0.036
ELYZA _{7b}	0.352/0.520	0.628/0.771	0.003/0.046	0.196/0.044	0.015/-0.004
GPT-3.5	0.369/0.489	0.528/0.570	-0.013/-0.022	0.255/0.064	0.389/0.385
GPT-4	0.325/0.433	0.583/0.612	0.028/0.073	0.417/0.113	0.776/0.811
<i>Fine-tuned LLMs</i>					
ELYZA _{7b}	0.638/0.638	0.692/0.694	0.240/0.235	0.379/0.280	0.684/0.740
Human	0.732/0.790	0.703/ 0.807	—	0.564/0.538	0.699/0.765

Experiment

Key Findings in a Nutshell

Experiment

Key Findings in a Nutshell

- **Fine-tuning encoder PLMs still perform well in AdTEC**
 - Achieved the highest score in three out of five tasks

Experiment

Key Findings in a Nutshell

- **Fine-tuning encoder PLMs still perform well in AdTEC**
 - Achieved the highest score in three out of five tasks
- **Humans surpassed PLMs in Ad Acceptability and Consistency tasks**
 - In the Ad Acceptability task, models miss semantic issues such as redundancy or contradiction
 - In the Ad Consistency task, models struggled when LPs omit meta expressions like “official”

Experiment

Key Findings in a Nutshell

- **Fine-tuning encoder PLMs still perform well in AdTEC**
 - Achieved the highest score in three out of five tasks
- **Humans surpassed PLMs in Ad Acceptability and Consistency tasks**
 - In the Ad Acceptability task, models miss semantic issues such as redundancy or contradiction
 - In the Ad Consistency task, models struggled when LPs omit meta expressions like “official”
- **GPT-4 and fine-tuned models beat human performance in Ad Similarity**
 - PLMs can better understand nuanced relationships between ad content

Experiment

Key Findings in a Nutshell

- **Fine-tuning encoder PLMs still perform well in AdTEC**
 - Achieved the highest score in three out of five tasks
- **Humans surpassed PLMs in Ad Acceptability and Consistency tasks**
 - In the Ad Acceptability task, models miss semantic issues such as redundancy or contradiction
 - In the Ad Consistency task, models struggled when LPs omit meta expressions like “official”
- **GPT-4 and fine-tuned models beat human performance in Ad Similarity**
 - PLMs can better understand nuanced relationships between ad content
- **Open-source LLMs struggled with semantic similarity and numerical prediction tasks**
 - In the Ad Performance Estimation task, all open-sourced LLMs show nearly no correlation with quality scores
 - An insufficient understanding of quantitative reasoning

Experiment

Additional Error Analysis



- Examples of integrated gradient visualization [15] with Tohoku BERT model's outputs in the AD SIMILARITY task
 - The difference in attention for small (top) and large (bottom) gaps between ground truth and predicted labels
 - Red is negative influence and green is positive influence on the predictions
- PLMs focus on surface features (e.g. numerical features) and miss semantic similarity and paraphrase

Conclusion

- We **organized advertising workflows** based on real-world AdOps
- We **designed five related tasks, created datasets to evaluate ad text quality** based on the workflow, and **packaged them as a benchmark called AdTEC**.
- We **validated the performance of existing PLMs and human evaluators** on our benchmark.
- We **demonstrated that our benchmark is challenging** through our experiments, as humans outperformed PLMs in two tasks in the results.
- We **analyzed the results and found that the following area still remain challenging** for PLMs, which we identified as areas for future work
 - Understanding the hidden intentions behind words
 - Understanding the unique characteristics of advertising
 - Understanding the comprehension of numerical/quantitative information

References

1. PwC group, **Changes in digital advertising spend**, 2025
<https://www.pwc.com/gx/en/industries/tmt/media/outlook/changes-in-digital-advertising-spend.html>
2. Dentsu group, **Comprehensive Analysis of Japan's 2024 Digital Ad Spending (In Japanese)**, 2025
<https://www.dentsu.co.jp/news/release/2025/0312-010858.html>
3. Hughes et al., **Generating Better Search Engine Text Advertisements with Deep Reinforcement Learning**, KDD 2019
4. Kamigaito et al., **An Empirical Study of Generating Texts for Search Engine Advertising**, NAACL 2021 Industry Track
5. Golobokov et al., **DeepGen: Diverse Search Ad Generation and Real-Time Customization**, EMNLP 2022 Demo Track
6. Mita et al., **Striking Gold in Advertising: Standardization and Exploration of Ad Text Generation**, ACL 2024
7. Murakami et al., **Natural Language Generation for Advertising: A Survey**, arXiv 2023
8. Appier, **AdCreative.ai**, <https://www.adcreative.ai/>
9. Adriel, **AdGen AI**, <https://www.adgenai.com/>
10. JACQUARD, **JACQUARD**, <https://www.jacquard.com/>
11. CyberAgent, **KiwamiTD**, <https://www.cyberagent.co.jp/news/detail/id=24670>
12. Dentsu, **Mugen AI**, <https://www.dentsudigital.co.jp/services/AI/mugen-ai>
13. Hakuholdo, **CREATIVE BLOOM TEXT Ads**, <https://www.hakuholdo-holdings.co.jp/news/corporate/2024/12/5173.html>
14. Murakami et al., **Aspect-based Analysis of Advertising Appeals for Search Engine Advertising**, NAACL 2022 Industry Track
15. Sundararajan et al., **Axiomatic Attribution for Deep Networks**, ICML 2017

Thank you for listening!

✎ 2408.05906



CyberAgentAILab/AdTEC



cyberagentailab.github.io/AdTEC

Come visit our poster!

Wednesday, April 30, 11:00—12:30

Session B (R&E: Resources and Evaluation), Hall 3